Basic Network Configuration: Setting up and Troubleshooting

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Table of Contents

1.0 Introduction
2.0 Aim and objectives
2.1 Tools
3.0 Methodology
3.1 Pinging the OWASP BWA
3.2 Tracing the path to the OWASP Application
3.3 Viewing Active Connections to OWASP VM
3.4 Transport Layer Protocols (TCP and UDP)
3.5 Discovering Mac Address with ARP
3.6 Capturing Network Traffic with Wireshark
4.0 Results and Analysis
4.1 OSI Layers and Functions with TCP/IP Mapping
5.0 Conclusion and Recommendation

Executive Summary

This lab focused on exploring the OSI and TCP/IP models using the IP address of the OWASP Broken Web Application virtual machine. Key activities include testing host reachability , verifying host connection, confirming connection type between host (tcp or udp) and applying packet filters to narrow down results. The lab revealed that successful ping to a host doesn't guarantee active connection. An icmp reply from the ping process is a function of layer 3 while tcp or udp connection at layer 4 is a function of 3-ways handshakes.

1.0 Introduction

The OSI (Open Systems Interconnection) model is a 7-layer conceptual framework that explains how data moves from one device to another over a network. Each layer has a specific function, from physical transmission of bits up to application-level communication, allowing network processes to be standardized, understood, and troubleshot layer by layer.

The TCP/IP model is a 4-layer practical networking model used on the Internet that defines how data is packaged, addressed, transmitted, routed, and received. It focuses on end-to-end communication using protocols like IP for routing and TCP/UDP for transport, enabling reliable and scalable data exchange across interconnected networks.

2.0 Aim and objectives

The aim of this lab is to explore the OSI model and TCP/IP suite and study their operational capabilities using a target website(owaspbwa).

The Objectives of this lab is to

- Understand the OSI and TCP models and their functions
- Use essential network tools to explore and analyse network traffic
- Apply knowledge of TCP/IP layers in real world scenarios using OWASP broken web Application IP

2.1 Tools

- Kali Linux
- OWASP BWA
- Wireshark
- Tshark

3.0 Methodology

3.1 Pinging the OWASP BWA

- The Linux terminal was opened and the following ping command was run
- **Ping -c 4 192.168.56.102**
- An Icmp reply was received indicating that the target is alive, active, reachable and responding
- Ping is an operation of layer 3 verifying the reachability of an ip address on a network.

3.2 Tracing the path to the OWASP Application

- The Linux terminal was opened and the following command was run
- **Traceroute 192.168.56.102**
- The traceroute has **30 hops** maximum to reach the OWASP BWA machine
- Each hop in a traceroute represents a router or Layer 3 device that a packet passes through on its way to a destination.

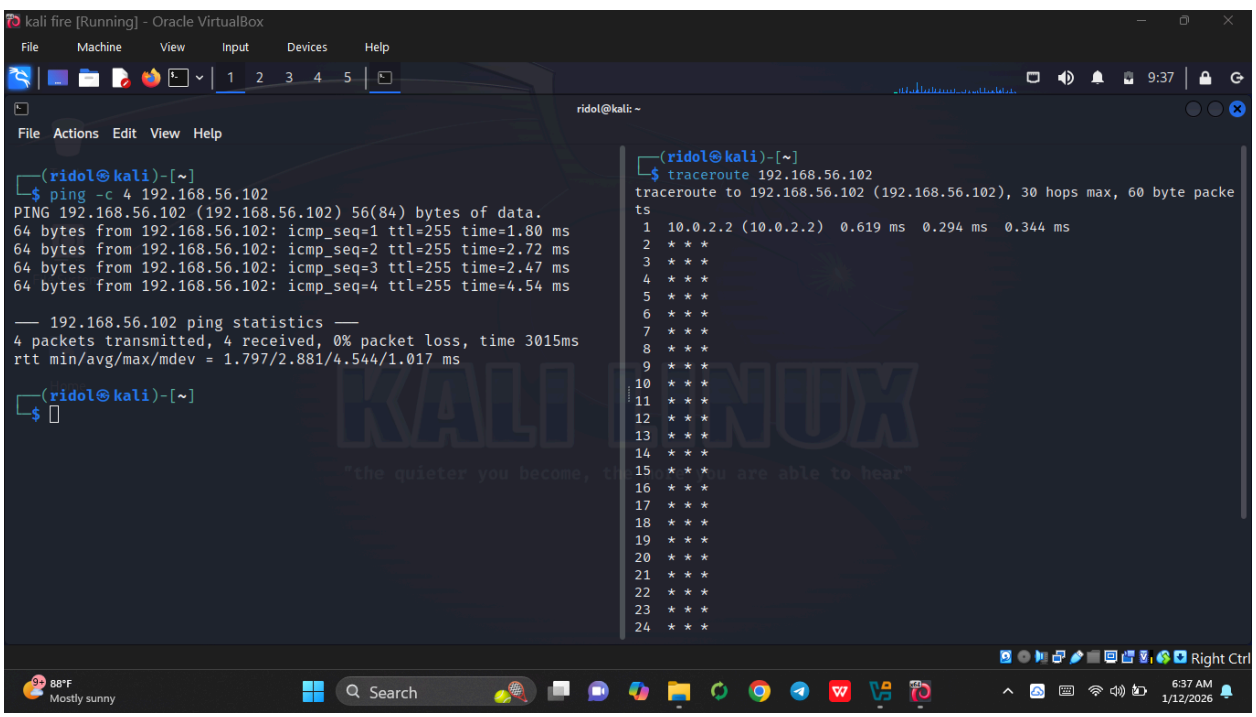

Fig 1:Evidence showing a successful ping and tranceroute of a target on Linux terminal

3.3 Viewing Active Connections to OWASP VM

- The Linux terminal was opened and the following command was run
- **netstat -an | grep 192.168.56.102** and **ss -an | grep 192.168.56.102**
- This showed an established **tcp connection** between the kali Linux and the owasp bwa machine **ip addresses** and the connection **port**
- This indicates a successful 3-ways handshakes on the transport layers **(SYN SYN/ACK ACK)**

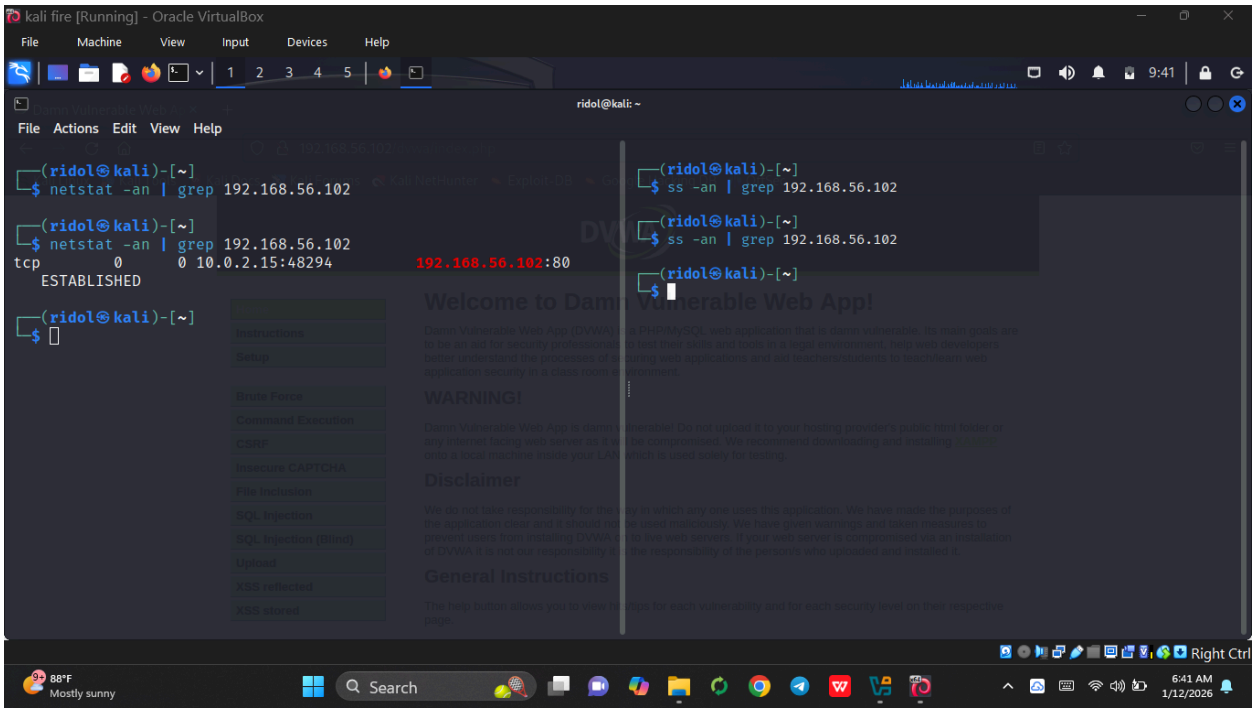

Fig 2: Evidence showing a successful tcp connection on Linux terminal

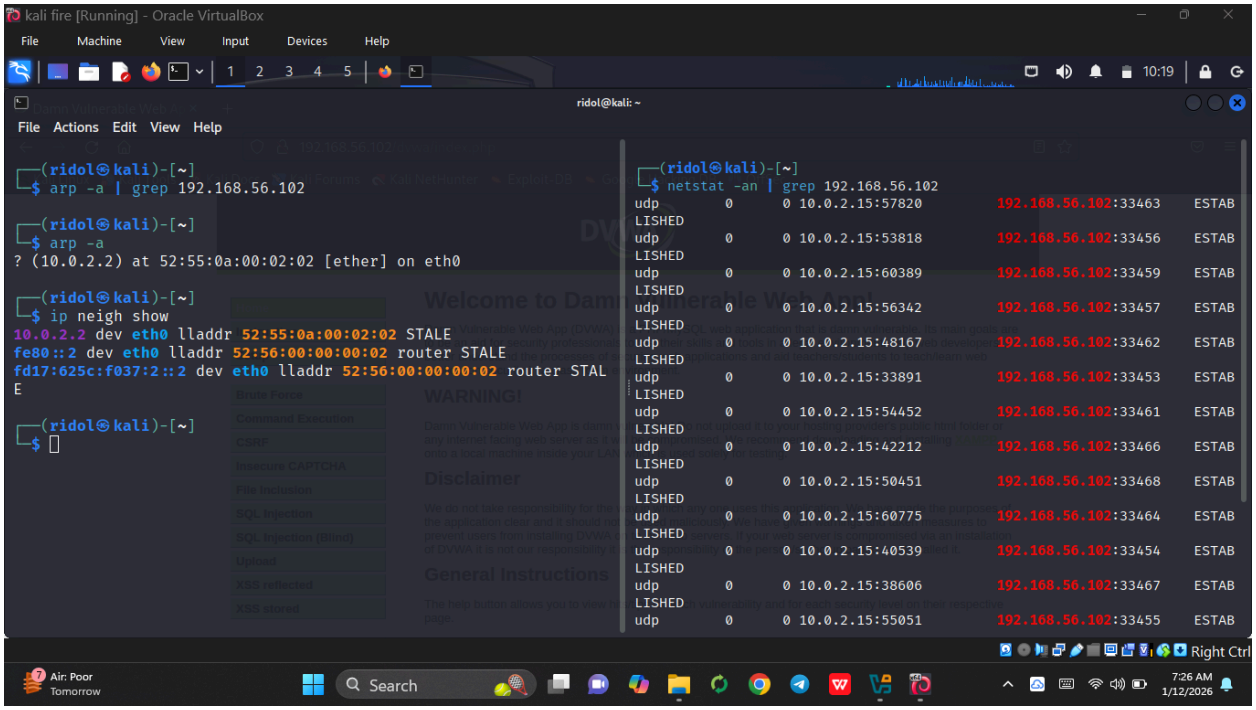

Fig 3: Evidence showing a successful tcp connection on Linux terminal

3.4 Transport Layer Protocols (TCP and UDP)

- The Linux terminal was opened and the following command was run
- **Sudo nmap -sT 192.168.56.102** and **Sudo nmap -sU 192.168.56.102**
- The network scan with the **-sT** argument returned all **opened ports specifically tcp** connections on the target IP network
- The network scan with the **-sU** argument returned all **UDP** connections on the target IP network
- The **UDP** connections are mostly sensitive to files carrying traffic.

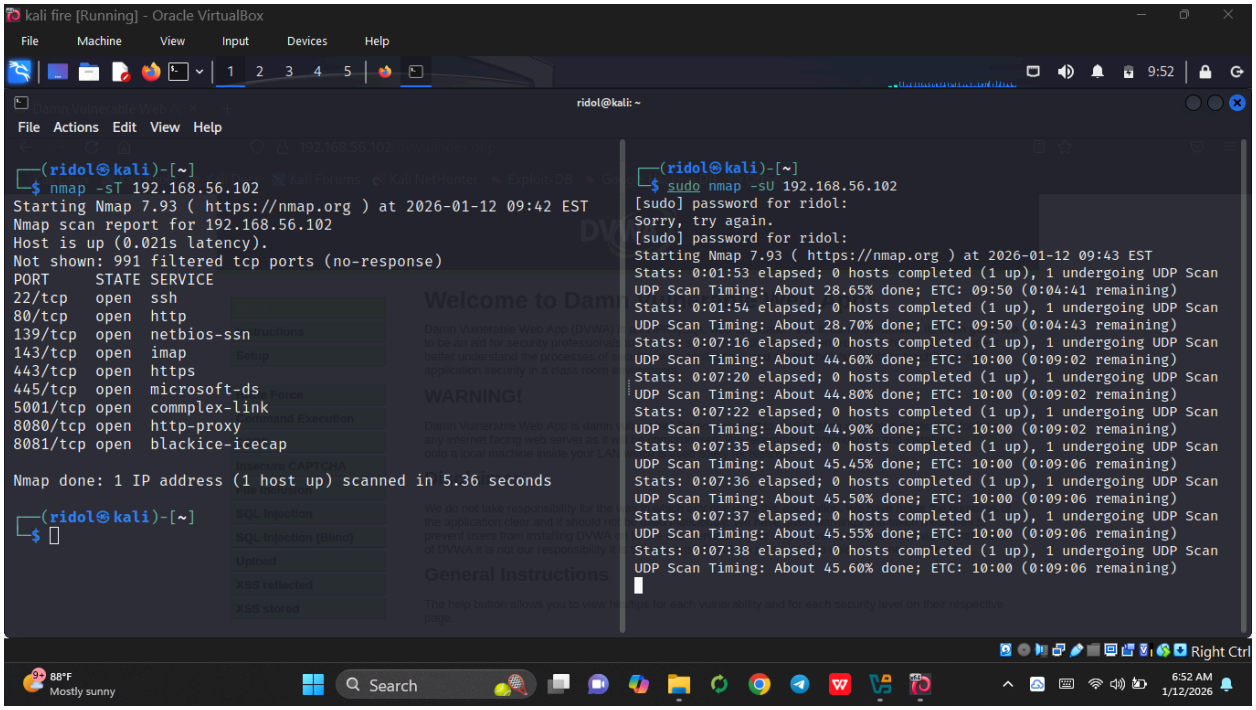

Fig 4: Evidence showing a successful tcp connection and udp traffic on Linux terminal

3.5 Discovering Mac Address with ARP

- The Linux terminal was opened and the following command was run
- **Arp -a grep 192.168.56.102** and **arp -a**
- The second command returned a MAC address associated to the Linux machine

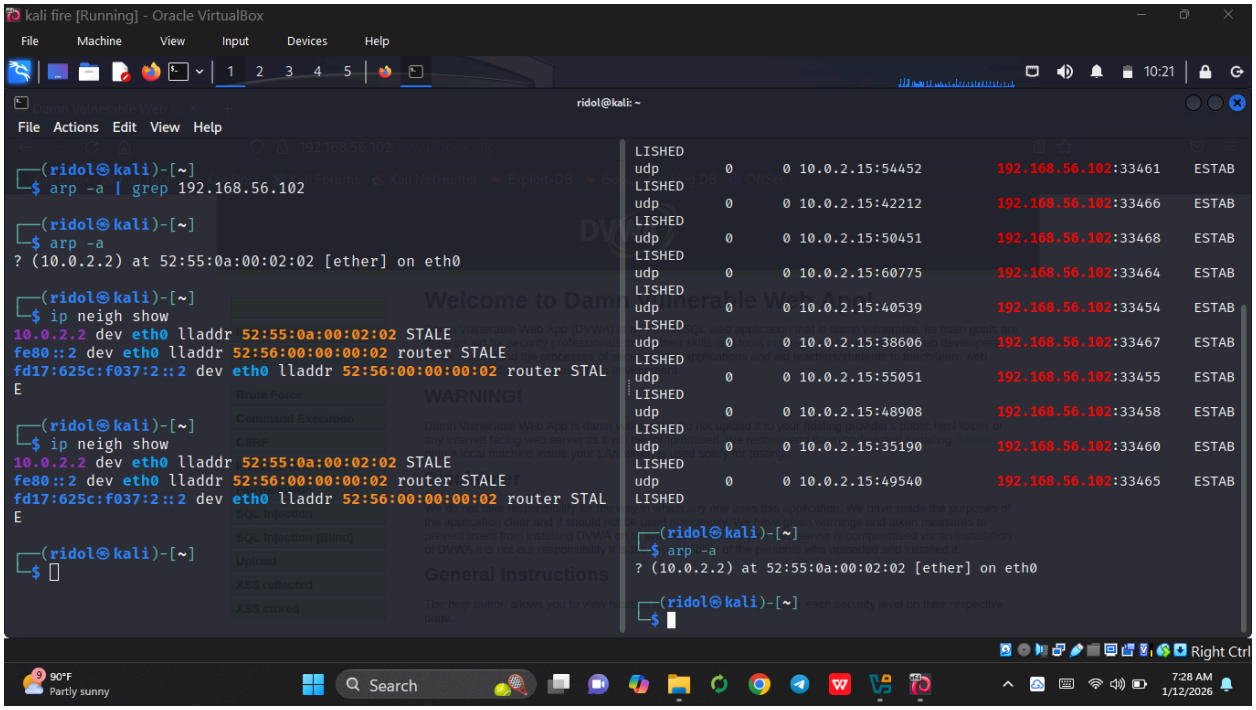

Fig 5: Evidence showing Mac address on Linux terminal

3.6 Capturing Network Traffic with Wireshark

- The wireshark was launched from the Linux terminal using the command **wireshark**
- Packet from owaspbwa was captured on the wireshark while some pages were visited on the owaspbwa web server
- A successful and complete tcp connection, specifically the 3 ways handshake was observed from the packet sequence.

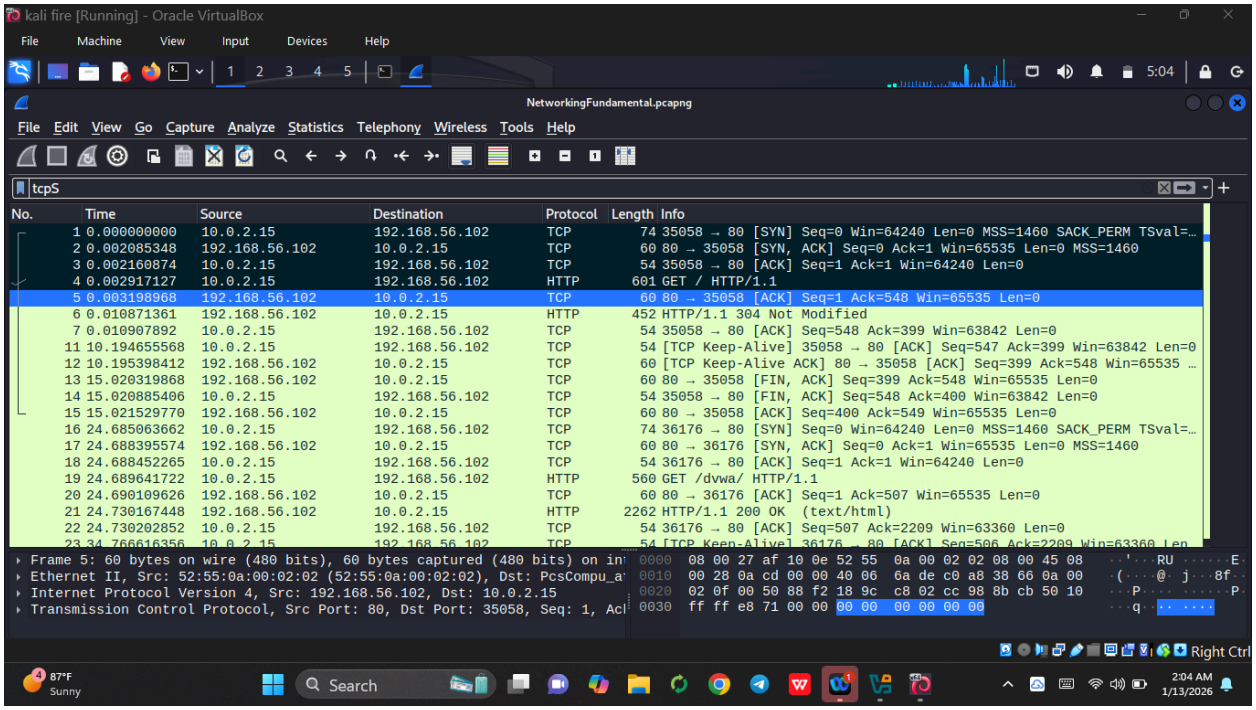

Fig 6: Evidence showing a successful 3-ways handshakes in the first 3 packets on wireshark interface

4.0 Results and Analysis

4.1 OSI Layers and Functions with TCP/IP Mapping

1. **Physical** – Transmits raw bits over the medium (cables, signals).

   TCP/IP: Network Access

2. **Data Link** – Framing, MAC addressing, error detection (Ethernet, ARP).

   TCP/IP: Network Access

3. **Network** – Logical addressing and routing (IP, routers).

   TCP/IP: Internet

4. **Transport** – End-to-end delivery, reliability, flow control (TCP/UDP).

   TCP/IP: Transport

5. **Session** – Manages sessions and connections.

   TCP/IP: Application

6. **Presentation** – Data format, encryption, compression.
   TCP/IP: Application
7. **Application** – User-facing network services (HTTP, FTP, SMTP).
   TCP/IP: Application

TCP is reliable because it establishes a connection, ensures packets are delivered in order, and retransmits lost data, but this adds overhead and makes it slower.

UDP is faster because it sends data without establishing a connection or checking delivery, but it is unreliable since packets can be lost, duplicated, or arrive out of order.

ARP (Address Resolution Protocol) is significant at the Data Link Layer because it maps an IP address to a physical MAC address. Since frames on a local network are delivered using MAC addresses, ARP allows a device to find the correct hardware address of the destination before sending data.

It contributes to successful communication by ensuring that packets meant for a specific IP are correctly encapsulated into frames and delivered to the right device on the local network.

5.0 Conclusion and Recommendation

The OSI model and TCP/IP suite plays an important role in users' interconnection on the internet and each layer has its own possible exploitable weaknesses if not properly configured or secured. The layer 1 and 2 of the OSI model were merged together to form the data-link layer of the TCP/IP suite and the layer 5,6 and 7 were merged to form the application layer of the TCP/IP suite.

Its important and highly recommended to understand the individual operations of each layer and how they influence each other throughout the encapsulation and de-encapsulation process, especially the **data-segment-packet-ethernet frame** chains with their headers.